DECENT COACHING CLASSES

IX CBSE

Mind Your Math - IX

ANSWER KEY & MARKING SCHEME

Time: 3 Hours **Maximum Marks: 80**

SECTION A (1 mark each)

Q.	1	2	3	4	5	6	7	8	9	10
Ans.	(a)	(c)	(a)	(b)	(c)	(a)	(c)	(c)	(c)	(c)

Q.	11	12	13	14	15	16	17	18	19	20
Ans.	(d)	(b)	(c)	(b)	(a)	(a)	(c)	(b)	(c)	(b)

Hints / working:

1. $(-3, 4) \to$ y-axis reflection $(3, 4) \to$ x-axis reflection $(3, -4)$.
2. $x^2 + 16 = 25 \Rightarrow x = \pm 3$.
3. $p(-3) = 0 \Rightarrow b = 3a$; $a + b = 8 \Rightarrow a = 2, b = 6$.
4. Intercepts $(2, 0)$ and $(0, -4)$; area $= \frac{1}{2} \times 2 \times 4 = 4$.
5. $(2 + \sqrt{3})^2 = 7 + 4\sqrt{3}$ is irrational; others equal 2, 3 and 0.3.
6. $81^{-3/4} = 3^{-3} = 1/27$; $(1/9)^{-1} = 9$; product $= 1/3$.
7. Non-terminating, non-repeating decimal lying between 0.333… and 0.5.
8. $7/80 = 0.0875$ (denominator $2^4 \times 5$).
9. Square: $a^2 + 1/a^2 = (a - 1/a)^2 + 2 = 16 + 2 = 18$.
10. $(x + y)^3 - (x - y)^3 = 2(3x^2y + y^3)$.
11. $x^2 + kx + 36 = (x \pm 6)^2 \Rightarrow k = \pm 12$.
12. $r^2 = 8^2 + 6^2 = 100 \Rightarrow r = 10$ cm.
13. $\angle A + \angle C = 180° \Rightarrow 6x = 180 \Rightarrow x = 30$.
14. Angle at the centre is twice the angle at the major arc: $140° \div 2 = 70°$.
15. $s = 21$; area $= \sqrt{(21 \times 8 \times 7 \times 6)} = 84$ cm^2.
16. Side 20 m; area $= (\sqrt{3}/4) \times 20^2 = 100\sqrt{3}$ m^2.
17. Prime or even $= \{2, 3, 4, 5, 6\} \Rightarrow 5/6$.
18. $b/(6 + b) = 2/5 \Rightarrow 5b = 12 + 2b \Rightarrow b = 4$.
19. $a + 4d = 22$, $a + 8d = 38 \Rightarrow d = 4$, $a = 6$; $a_{15} = 6 + 56 = 62$.
20. $a_{10} = S_{10} - S_9 = 230 - 189 = 41$.

SECTION B (2 marks each)

Q.21

$AB = \sqrt{[(2 + 2)^2 + (1 - 1)^2]} = 4$ [½]
$AC = \sqrt{[(0 + 2)^2 + (5 - 1)^2]} = \sqrt{20} = 2\sqrt{5}$ [½]
$BC = \sqrt{[(0 - 2)^2 + (5 - 1)^2]} = \sqrt{20} = 2\sqrt{5}$ [½]
$AC = BC \Rightarrow$ triangle ABC is isosceles. [½]

Q.22

Using (1, 5): $5 = m + c$; using (3, 11): $11 = 3m + c$. [1]
Subtracting: $2m = 6 \Rightarrow \mathbf{m = 3}$ [½]; $c = 5 - 3 = \mathbf{2}$ [½]

Q.23

Let $x = 0.3666…$ Then $10x = 3.666…$ and $100x = 36.666…$ [1]
$100x - 10x = 33 \Rightarrow 90x = 33 \Rightarrow \mathbf{x = 11/30}$ [1]

Q.24

$x^3 + y^3 = (x + y)^3 - 3xy(x + y)$ [1]

$= 5^3 - 3(6)(5) = 125 - 90 =$ **35** [1]

Q.25

Let ABCD be a cyclic parallelogram. Opposite angles of a parallelogram are equal: $\angle A = \angle C$. [½]

Opposite angles of a cyclic quadrilateral are supplementary: $\angle A + \angle C = 180°$. [½]

Hence $\angle A = \angle C = 90°$. [½]

A parallelogram with one angle 90° is a rectangle. [½]

SECTION C (3 marks each)

Q.26

$AB = \sqrt{(16 + 4)} = \sqrt{20}$, $BC = \sqrt{(4 + 16)} = \sqrt{20}$, $CD = \sqrt{(16 + 4)} = \sqrt{20}$, $DA = \sqrt{(4 + 16)} = \sqrt{20}$ [1½]

$AC = \sqrt{(4 + 36)} = \sqrt{40}$, $BD = \sqrt{(36 + 4)} = \sqrt{40}$ [1]

All sides equal and diagonals equal ⇒ ABCD is a square. [½]

Q.27

$(3 + \sqrt{2})/(3 - \sqrt{2}) \times (3 + \sqrt{2})/(3 + \sqrt{2}) = (3 + \sqrt{2})^2/(9 - 2) = (11 + 6\sqrt{2})/7$ [2]

$a = 11/7, b = 6/7$ [1]

OR $(7 + 3\sqrt{5})/(3 + \sqrt{5}) = (7 + 3\sqrt{5})(3 - \sqrt{5})/4 = (6 + 2\sqrt{5})/4$ [1]

$(7 - 3\sqrt{5})/(3 - \sqrt{5}) = (7 - 3\sqrt{5})(3 + \sqrt{5})/4 = (6 - 2\sqrt{5})/4$ [1]

Difference $= 4\sqrt{5}/4 =$ **$\sqrt{5}$** [1]

Q.28

$(2x + 1)^3 = 8x^3 + 12x^2 + 6x + 1$, so the expression $= (2x + 1)^3 - 8$ [1]

$= (2x + 1)^3 - 2^3 = (2x + 1 - 2)[(2x + 1)^2 + 2(2x + 1) + 4]$ [1]

$=$ **$(2x - 1)(4x^2 + 8x + 7)$** [1]

Q.29

Sides = 5k, 12k, 13k; 30k = 300 ⇒ k = 10; sides 50 m, 120 m, 130 m. [1]

s = 150 m; Area $= \sqrt{(150 \times 100 \times 30 \times 20)} = \sqrt{9{,}000{,}000} =$ **3000 m²** [1]

Cost $= 3000 \times 4 =$ **₹ 12,000** [1]

Q.30

Total outcomes = 30.

(i) Primes: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29 (10) ⇒ **1/3** [1]

(ii) Perfect squares: 1, 4, 9, 16, 25 (5) ⇒ 5/30 = **1/6** [1]

(iii) Multiples of 15: 15, 30 (2) ⇒ 2/30 = **1/15** [1]

Q.31

$a = 24, d = -3$. $S_n = (n/2)[48 + (n - 1)(-3)] = 78$ [1]

$n(51 - 3n) = 156 \Rightarrow n^2 - 17n + 52 = 0 \Rightarrow (n - 4)(n - 13) = 0 \Rightarrow$ **n = 4 or n = 13** [1]

Reason: the terms 5th to 13th (12, 9, 6, 3, 0, −3, −6, −9, −12) add up to 0, so the sum stays 78 up to the 13th term. [1]

SECTION D (5 marks each)

Q.32

Table of values: (0, 4), (3, 2), (6, 0) (also (−3, 6)). [1]

Correct plotting of at least 3 points on graph paper [1½] and the straight line drawn through them [1½].

(i) $x = 3 \Rightarrow$ **$y = 2$** [½] (ii) $y = 6 \Rightarrow$ **$x = -3$** [½]

(iii) Intercepts 6 and 4 ⇒ area $= ½ \times 6 \times 4 =$ **12 sq. units** [1]

Q.33

$x = 1/(3 - 2\sqrt{2}) = (3 + 2\sqrt{2})/(9 - 8) = 3 + 2\sqrt{2}$, so $1/x = 3 - 2\sqrt{2}$. [1]

(i) $x + 1/x =$ **6** [1]

(ii) $x^2 + 1/x^2 = 6^2 - 2 =$ **34** [1½]

(iii) $x^3 + 1/x^3 = 6^3 - 3(6) = 216 - 18 = \mathbf{198}$ [1½]

OR $a^2 + b^2 + c^2 = 9^2 - 2(26) = 81 - 52 = \mathbf{29}$ [1½]

$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) = 9(29 - 26) = \mathbf{27}$ [2]

$a^3 + b^3 + c^3 = 27 + 3(24) = \mathbf{99}$ [1½]

Q.34

OA = OB (radii) and O′A = O′B (radii), so both O and O′ lie on the perpendicular bisector of AB. Hence line OO′ is the perpendicular bisector of AB. [2]

In △OAO′: $OA^2 = 25$, $OO'^2 + O'A^2 = 16 + 9 = 25 \Rightarrow \angle AO'O = 90°$. [1½]

So O′A ⊥ OO′, i.e. O′ is the midpoint of AB (the point where OO′ meets AB). Therefore AO′ = 3 cm. [1]

AB = 2 × 3 = **6 cm** [½]

Q.35

In right △ABC: $AC^2 = 9^2 + 40^2 = 1681 \Rightarrow AC = 41$ m. [1]

Area of △ABC = ½ × 9 × 40 = 180 m². [1]

For △ACD: sides 41, 15, 28; s = 42 m. [1]

Area = $\sqrt{(42 \times 1 \times 27 \times 14)} = \sqrt{15876} = 126$ m². [1]

Area of field = 180 + 126 = **306 m²** [1]

OR s = 84 m; Area = $\sqrt{(84 \times 32 \times 28 \times 24)} = \mathbf{1344\ m^2}$ [3]

Altitude on 56 m: ½ × 56 × h = 1344 ⇒ **h = 48 m** [1]

Shortest altitude is on the longest side (60 m): h = 2 × 1344/60 = **44.8 m** [1]

SECTION E (4 marks each)

Q.36 Case Study 1

(i) AB = 6 units = **60 m** [1]

(ii) AC = $\sqrt{(6^2 + 8^2)} = 10$ units = **100 m** [1]

(iii) Midpoint of AC = ((1 + 7)/2, (2 + 10)/2) = **(4, 6)** [1]

(iv) **D = (1, 10)** [1]

Q.37 Case Study 2

The weekly savings form an AP: 50, 70, 90, … with a = 50, d = 20.

(i) $a_{12} = 50 + 11 \times 20 = \mathbf{₹\ 270}$ [1]

(ii) $50 + (n - 1)20 = 410 \Rightarrow n - 1 = 18 \Rightarrow$ **19th week** [1]

(iii) $S_{15} = (15/2)[2(50) + 14(20)] = (15/2)(380) = \mathbf{₹\ 2850}$ [2]

Q.38 Case Study 3

(i) P(red) = **3/8** [1]

(ii) P(not green) = 1 − 2/8 = **3/4** [1]

(iii) Even numbers 2, 4, 6, 8 ⇒ P = 4/8 = **1/2** [1]

(iv) Favourable sectors {2, 3, 5, 7} ∪ {4, 5} = {2, 3, 4, 5, 7} ⇒ P = **5/8** [1]

— End of Answer Key —